



**Linux 101: It's Everywhere
(Except Your Laptop)**

The hobby that wasn't supposed to be big



Helsinki, 1991 • Linus, 21, cold dorm

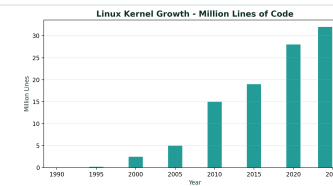
ORIGINAL POST

"I'm doing a (free) operating system (just a hobby, won't be big and professional like GNU)..."

Wanted to call it **FREAX** — Free + Freak + X. FTP admin created folder /pub/OS/Linux without permission. Name stuck forever.

10K → 30M LINES

10k **30M+** **3000×**
V0.01 1991 TODAY GROWTH



The biggest flame war in CS history

1992

Prof. Andrew Tanenbaum posts:

"Linux is obsolete. Microkernels are the future. Monolithic kernels belong in the past."

Reply

Linus fires back:

"Your job is being a professor... that's one hell of a good excuse for some of the brain-damages of MINIX."

- 1,200+ messages. Still taught in OS classes today.

Today

Who won?

Theory vs working code. Students love this debate because it's about getting sh*t shipped.

SCOREBOARD 2026

"Academic theory predicted MINIX. The world ran Linux."

LINUX

**100% of Top 500
Supercomputers**

MINIX

**Hidden inside every Intel chip
as Management Engine**

Speaker note: Great example — Linus built something ugly that worked, Tanenbaum built something elegant that didn't ship. Which approach wins interviews?

1993 • CANBERRA NATIONAL ZOO

Why is the mascot a fat penguin?

Linus was bitten by a Little Penguin. Got a slight fever on the flight home. Thought it was funny and “oddly cute” — not scary.

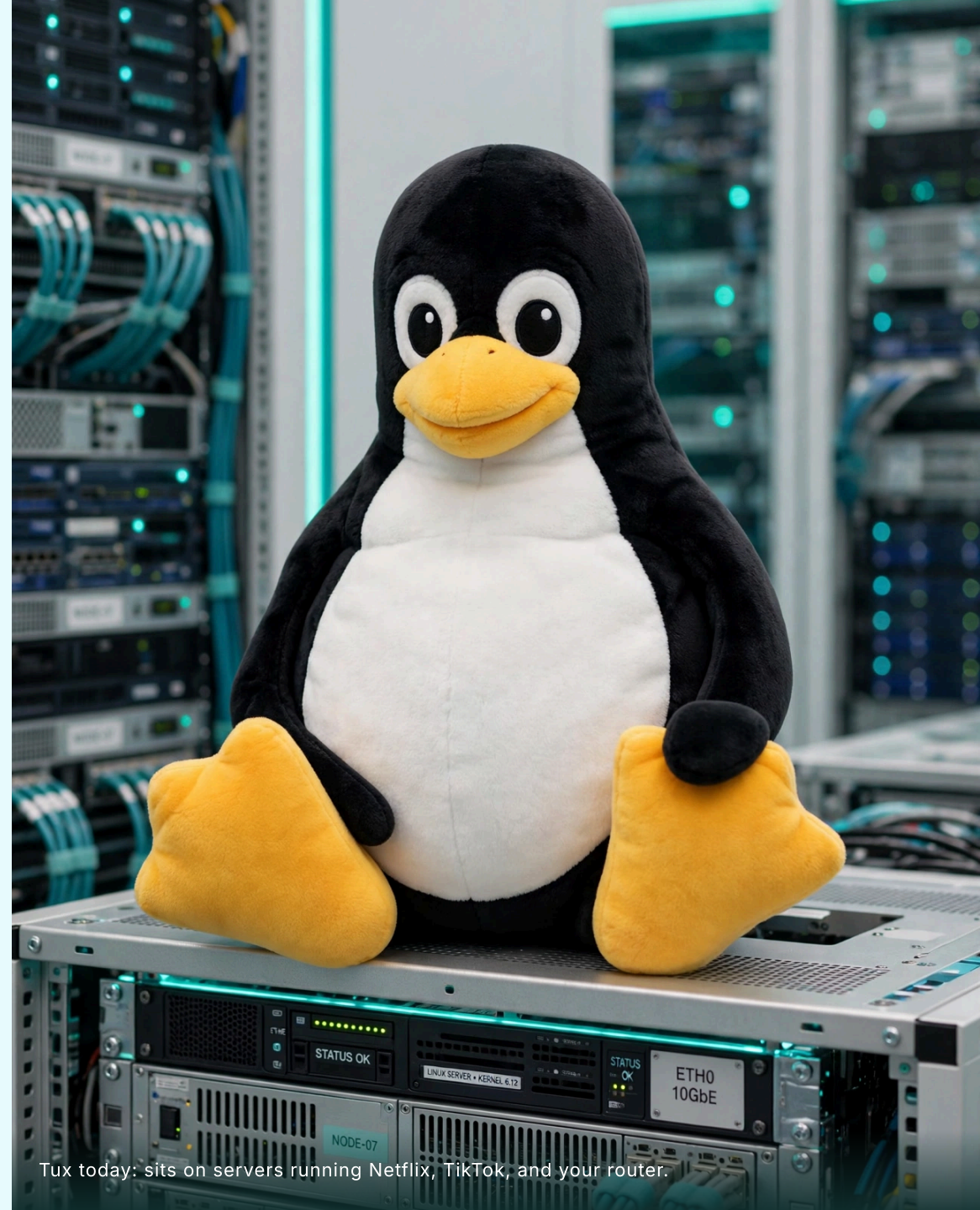
Needed a logo in 1996. Artist **Larry Ewing** drew **Tux** — short for *Torvalds UniX* — using GIMP. First open-source mascot made with open-source tools.

Linus rule: Tux should look like he just ate way too much and is happily sitting down. No aggressive eagle. No corporate polish.

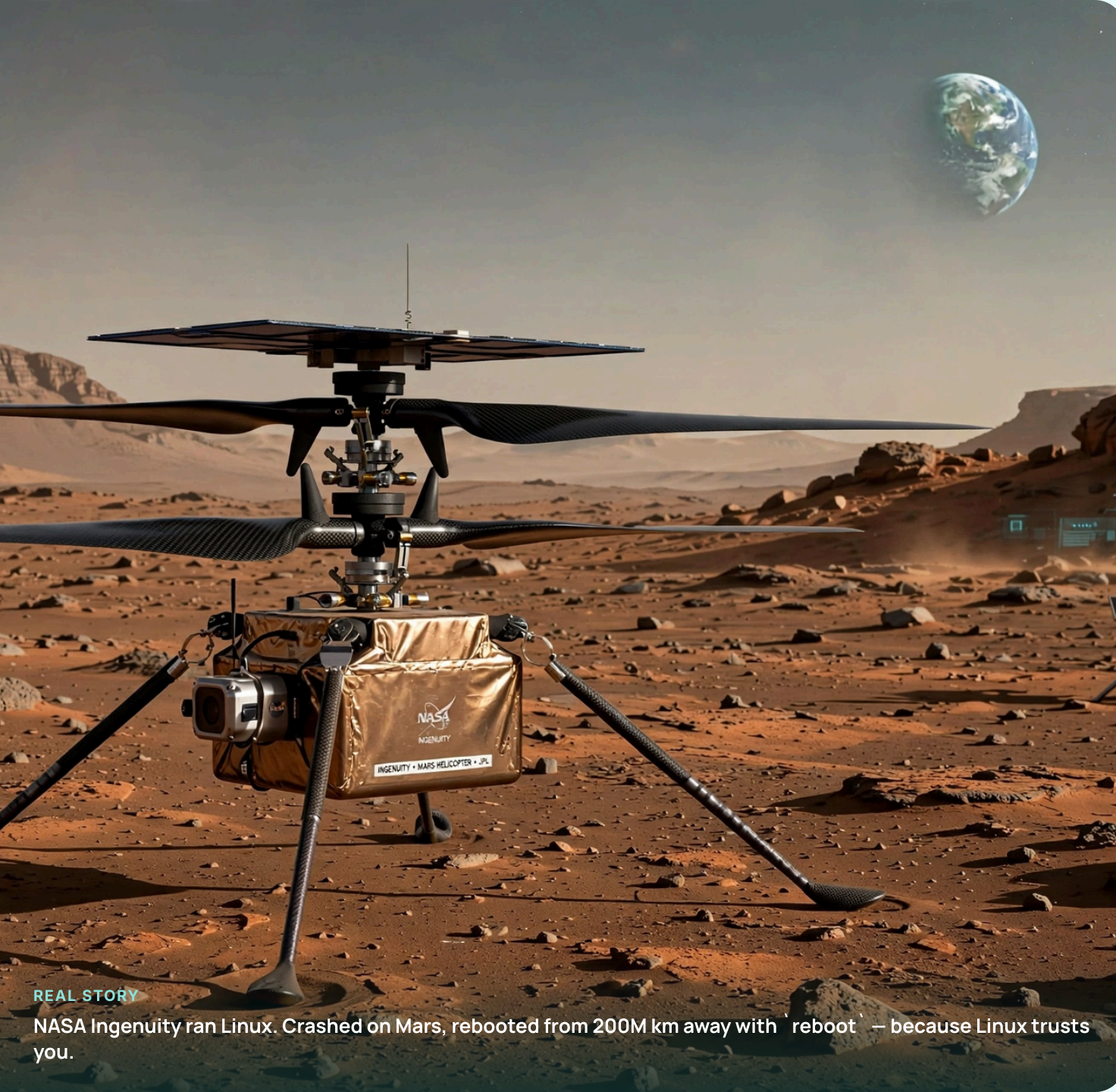
🐧 Little Penguin • *Eudyptula minor*

🎨 Larry Ewing • GIMP 0.54

```
user@lab:~$ sudo apt install cowsay -y
user@lab:~$ cowsay -f tux "Hello class"
→ Try it live. Breaks the ice instantly.
```



Tux today: sits on servers running Netflix, TikTok, and your router.



REAL STORY

NASA Ingenuity ran Linux. Crashed on Mars, rebooted from 200M km away with 'reboot' — because Linux trusts you.

Where is Linux RIGHT NOW?



70%+ Phones = Linux

Android is Linux kernel. You already use it 20x/day.



96% of top web

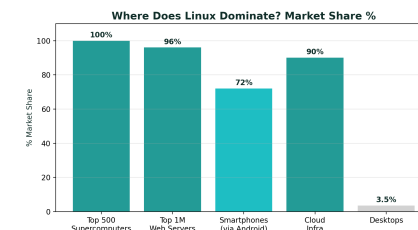
Netflix, TikTok, Facebook, Google — all Linux servers.



100% Supercomputers + Space

Top 500 all Linux. Even your fridge & router probably Linux.

LINUX DOMINANCE — MARKET SHARE %



From “cancer” to top contributor

2001 • ENEMY ERA

“Linux is a cancer that attaches itself in an intellectual property sense to everything it touches.”

— Steve Ballmer, Microsoft CEO

FUD campaigns, lawsuits, “Get the Facts” anti-Linux ads.

MICROSOFT STANCE

Closed, Windows-only

LINUX VIEW

Threat to business model

2024-2026 • JOINED THEM

Microsoft is now Top 5 contributor to the Linux kernel.

Azure runs more Linux than Windows. They ship Linux *inside* Windows via WSL2. VS Code, Typescript, GitHub all run on Linux infra.

AZURE

~60% Linux VMs

WSL

Linux kernel in Windows 11



Lesson: Build something so good your biggest enemy has to join you. Openness won.

ASK STUDENTS: WHAT DID LINUX DO RIGHT?

No one owns it — and that's the point

 LICENSE: GPL V2

Free forever.

Anyone can use, modify, sell, and share. No royalties. No permission needed from Linus.

- ✓ **Google took Linux, made Android** — didn't pay Linus a dollar. That's legal.
- ✓ **Windows is closed.** You can't see the recipe. Linux is open cookbook.
- ✓ **Result:** Most installed OS on Earth, hidden in plain sight.

HOW IT'S BUILT

5,000+

DEVELOPERS PER RELEASE

500+

COMPANIES • INTEL, GOOGLE, META,
MICROSOFT

13 yrs

YOUNGEST PATCH AUTHOR EVER MERGED

 Anyone can email a patch to LKML mailing list.

 LINUS STILL MERGES

Benevolent dictator model

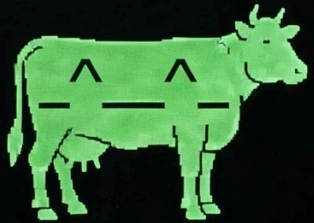
No corporation controls the kernel. Linus personally reviews and yells at bad code. Brutal, but fair.

-  Famous for blunt code reviews — quality over politeness
-  Trust built on transparency, not NDAs
-  Lesson: If you can't be bought, you can't be killed

 Open vs Closed — ask class which they'd trust for Mars?

Try this NOW – live terminal magic

```
user@devbox:~$ cowsay "Keep calm and sudo on"
```



Keep calm and sudo on!

```
user@devbox:~$ sl
```



```
user@devbox:~$ _
```

user@devbox:~\$ keeps it fun

cowsay + sl demo

ZERO-FEAR STARTER PACK

whoami # who are you?

uname -a # prove you're on Linux

history | tail # what you just did

\$ Let them run it. It breaks the ice.

MAKE THEM SMILE

sl typo ls → steam train chugs across screen

cowsay + fortune fortune | cowsay -f tux

cmatrix Matrix rain — instant wow

Linux assumes you know what you're doing. Respectful OS. Windows says "you can't." Linux says "are you sure? Okay."

Mythbusters – quick-fire trivia



TRUST TEST

rm -rf / can erase everything

Linux assumes you know what you're doing. No "Are you really sure?" pop-up. Respectful, not nanny.

🏆 Sticker if you explain sudo



HOLLYWOOD

90% of VFX rendered on Linux

Avatar, Avengers, Toy Story — all rendered on Linux farms. First Linux had zero GUI, just terminal.

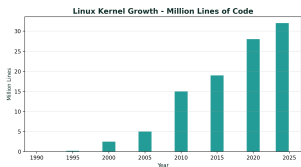
🖥️ No GUI in 1991

📄 Now 30M lines

GROWTH 3000x • PROOF

10k → 30M+ lines of kernel

From Linus solo to 5k contributors. Same kernel that runs your phone and Mars helicopter.



DON'T PICK A SIDE

vi vs emacs — 40-year holy war

Ask "which editor?" on Day 1 and you'll start a religion debate. Use nano for week 1, then choose. Pro tip: :wq saves in vi.

💡 Rapid-fire quiz — give stickers for correct exits



BONUS MYTH

"Linux is hard" — myth

If you can use Android, you already use Linux. The hard part was Windows telling you "you can't."

✓ 3% elite = you

10k
1991 V0.01

30M+
2026

You have the same tools as Linus did in 1991

"Windows says you can't. Linux says are you sure? Okay."

Next class: File system, permissions, your first bash script

Lab club: Wed 4pm • Open Source • Linux Club

Install it. Break it. Fix it. That's learning → `sudo apt update && curiosity`

THANK YOU • Q&A • STICKERS FOR BEST 'COWSAY' JOKE